

JAYADEEP GOPINATH

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PROFESSIONAL SUMMARY

MS Computer Science student at Illinois Tech building resilient distributed systems and production AI infrastructure in Go, Python, and Node.js. Shipped two flagships: an agentic RAG server over pgvector exposed via Model Context Protocol (MCP), and a multi-channel notification platform sustaining 170 req/s with p50 17ms in k6 load tests. Strong fundamentals in DSA, system design, and SDLC. Seeking Fall 2026 SWE / Backend / AI internships — F-1 CPT eligible (no H-1B sponsorship required).

TECHNICAL SKILLS

Programming Languages: Java, Python (primary), Go, C++, C#, JavaScript, C, TypeScript, PHP

Backend & APIs: FastAPI, Flask, Node.js, Express.js, gRPC, REST APIs, Model Context Protocol (MCP)

AI & ML: PyTorch, scikit-learn, LangChain, OpenAI, Anthropic, HuggingFace, sentence-transformers, RAGAS

Databases: PostgreSQL, pgvector (HNSW), MongoDB, MySQL, Redis — schema design, indexing, connection pooling

Cloud & Infrastructure: Docker, docker-compose, GitHub Actions CI/CD, AWS (EC2, S3, RDS), GCP, Linux, Prometheus, Grafana

CS Fundamentals & Testing: Data Structures & Algorithms, Distributed Systems, System Design, OS, pytest, go test, k6 load testing

EDUCATION

Illinois Institute of Technology — M.S. Computer Science

Chicago, IL | Aug 2025 – May 2027

Relevant Coursework: Advanced Database Organization, Computer Networks, Intro to Information Security

Visvesvaraya Technological University — B.E. Computer Science

Bengaluru, India | Dec 2020 – Jul 2024

Relevant Coursework: DSA, OOP, AI/ML, Algorithm Design, Big Data Analytics, Embedded Systems

PROFESSIONAL EXPERIENCE

Software Engineer Intern — Techfortune (Bengaluru, India)

August 2023 – October 2023

- Developed Python automated test scripts using FastAPI and Git to validate ML model predictions and backend API responses, covering **300+ test data records**.
- Refactored legacy Python codebase, standardized naming conventions, removed dead code, reducing average test execution time by **~40%**.
- Designed and optimized SQL schemas with composite indexes and views to improve query performance for test reporting dashboards.
- Documented test plans, findings, and validation reports; collaborated with a 4-engineer cross-functional team across SDLC phases, reducing handoff delays by 20%.

PROJECTS

Agentic-RAG-MCP — Agentic RAG via Model Context Protocol

2026

Python, MCP, pgvector (HNSW), FastAPI, OpenAI/Anthropic, sentence-transformers, Docker, GitHub Actions |

github.com/JayadeepGopinath/Agentic-RAG-MCP

- Built an MCP server (stdio + SSE) exposing an agentic RAG pipeline installable in Claude Desktop, Cursor, Cline, Continue, and Zed.
- Implemented **hybrid retrieval** (pgvector + BM25 fused with Reciprocal Rank Fusion) + cross-encoder reranker; multi-step agentic loop with planner, query expansion, and citation injection.
- RAGAS-style evaluation pipeline (faithfulness, answer relevance, context precision/recall) on a 20-question golden set; verified end-to-end at **p50 latency 293ms**, 24 chunks indexed across 4 docs.

Notification-system-go — Distributed Multi-Channel Notification Service

2026

Go, Redis Streams, PostgreSQL, Docker, Prometheus, Grafana, GitHub Actions | github.com/JayadeepGopinath/Notification-system-go

- Built REST API + worker pool with Redis Streams consumer groups for **at-least-once delivery**; sustained **170 req/s with p50 17ms** in k6 load tests (40,866 requests); identified Postgres pool saturation as next optimization target.
- Implemented exponential-backoff retries with dead-letter queue, idempotency via client-supplied keys, and per-channel token-bucket rate limiting.
- Multi-stage Docker build + docker-compose stack (Postgres + Redis + API + 3 workers + Prometheus + Grafana); GitHub Actions CI green; MIT-licensed.

CodePulse — Distributed Code Execution & Automated Test Validation

2025 – Present

Node.js, React, Socket.io, Docker, Redis, Python | github.com/JayadeepGopinath/codepulse

- Sandboxed Docker code execution with CPU/memory resource limits; Python pipelines validating code correctness across 5 languages with input sanitization and rate limiting.
- RESTful APIs with JWT auth sustaining **500+ concurrent sessions**; Redis Pub/Sub for real-time state sync; MongoDB compound indexes for **sub-50ms query performance**.

Naari Shakthi — AI-Powered Healthcare Platform

Govt. of Karnataka (KSCST) Recognized

React, Node.js, Python Flask, MongoDB, REST APIs | github.com/JayadeepGopinath/naari-shakthi

- Integrated Python ML service via REST APIs delivering real-time health risk predictions; validated outputs against clinical benchmarks with automated assertion checks.
- Implemented role-based access control, input validation, and rate-limited API endpoints; documented full system architecture for KSCST institutional deployment. **IEEE-published research** based on this project (IJTRE Vol. II, Jan 2024).

PUBLICATIONS

Jayadeep G, Dr. Mohan B R — “Smart Sanitary Pad and Medicine Vending Machine with Automated Dispensing, Disposal, and Notification System,” *IEEE Xplore Digital Library* (IJTRE, Vol. II Issue 5, Jan 2024, ISSN 2347-4718).